EpistemoBrain: A Seven-Scientist Council with Dual-Graph Evidence Governance for Auditable Scientific Blindspot Discovery

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Abstract

Scientific literature review is one of the highest-value applications of large language model (LLM) agents, yet also one of the most dangerous: a fluent, well-cited summary can mask a causal claim built on cross-sectional correlation, six papers that share a single dataset, or a dissent that was quietly flattened by a summarizer. We propose **EpistemoBrain**, an agentic framework that organizes **seven AI scientists with distinct specializations** into a formal council protocol – independent precommitment, professional acquisition, evidence exchange, cross- examination, blindspot bounty, joint modeling, and final rejudgment – to review a contested empirical question and produce an *auditable controversy evidence map* in which conclusions and limitations sit side by side, every claim is traceable to its source, and no dissent is deleted.

EpistemoBrain treats memory as the public cognitive layer of a scientific community rather than a private per-agent cache: three layers (private process memory, collective executive memory, and a formal evidence graph) are governed by a *dual-graph* architecture in which the process graph records how scientists worked while the evidence graph records what the community now knows. All formal changes pass through an append-only Scientific Event Ledger, a six-stage Evidence Gate, and a single Graph Projector whose write privileges are enforced at the database level. The protocol forbids majority voting as a way of deciding scientific truth: outcomes are expressed as conditional consensus, and genuine disagreement survives as named Dissent Certificates.

We evaluate EpistemoBrain on the **Foresight-Blindspot** benchmark, a time-sliced evaluation protocol for blindspot discovery in computational social science, and report a controlled scripted-case comparison of five baselines (single-agent deep research, fixed multi-agent debate, council with linear context, council with executive memory but no evidence engine, and the full system) plus a six-variant ablation ladder in which one capability is removed at a time. The comparison confirms that blindspot coverage is driven by the council's seat coverage – a lone researcher misses six of seven specialist blindspots – and the ablations show that removing the adversarial falsifier or the evidence auditor each costs a gold blindspot, while removing lineage tracking inflates the system's own evidence-independence report. A real-model run shows that keyword-based blindspot recall fails on free-form text and that a semantic judge repairs the measurement. Repeated runs then reveal a stable precision–recall trade-off rather than variance noise: the full system's governance achieves perfect blindspot precision at the cost of the lowest recall (two of seven gold concepts), while every simplified variant – including a lone researcher – finds nearly all gold concepts with lower precision; removing precommitment reproduces recall 1.0 in every run. End-to-end verification on real research tasks is reported alongside.

1 Introduction

LLM agents have made deep research practical: a user asks a question and an agent retrieves, reads, and synthesizes literature into a coherent answer (Yao et al., 2023; Lewis et al., 2020). For contested scientific questions, however, the standard deep-research pipeline inherits three failure modes that a fluency-optimized summary cannot fix. First, a single agent that reads a paper tends to see only its own interpretation; a measurement scientist's concern about self-report bias is invisible to a pipeline that has no measurement scientist. Second, agreement is not evidence: if seven agents read the same erroneous abstract, the same retracted paper, or the same misattributed citation, the system can package a *shared error* as a consensus that looks verified because it was repeated (Ioannidis, 2005; Open Science Collaboration, 2015). Third, a summarizer's final "everyone basically agrees" step flattens exactly the information a contested review must preserve: what the dissenting scientist opposed, and which evidence would change their mind. These are not prompt-tuning problems; they are structural questions about how a scientific review should be organized.

Computational social science makes the stakes concrete. The reproducibility crisis has shown that a large share of published findings do not replicate (Open Science Collaboration, 2015), that publication bias inflates apparent effect sizes (Fanelli, 2010), and that researcher degrees of freedom multiply false positives (Gelman and Loken, 2013; Nosek et al., 2018). In the specific domain this work targets – digital behavior, social media, and mental health – cross-sectional designs are routinely reported as causal, self-report measures dominate, and a small number of datasets support a large fraction of the literature (Orben and Przybylski, 2019; Valkenburg et al., 2022). An evidence reviewer for this domain needs more than an accurate summary: it needs to *audit* claims against their sources, to *count independent evidence* rather than papers, and to *keep the dissent* visible.

We propose **EpistemoBrain**, the organizing brain of **Poliscope**, a deep research agent that assembles **seven AI scientists** – theory builder, causal scientist, measurement and construct scientist, statistics and replication scientist, boundary and context scientist, adversarial falsifier, and evidence and provenance auditor – into a seven-round council protocol. The design starts from the opposite direction of "seven chatbots plus a summarizer": we first decide what organizational structure and memory mechanisms a reliable controversy review requires, then place the roles inside that structure. The architecture rests on four pillars: (i) all seven seats participate in every task, with cost controlled by speaking budgets, semantic deduplication, and shared retrieval caches rather than by dropping seats; (ii) dissent is preserved by construction – majority voting is forbidden, outcomes are conditional consensus, and dissents become named certificates; (iii) process and evidence are stored in separate graphs governed by an append-only event ledger and a single, database-enforced writer; and (iv) memory is a layered public cognitive layer – private per-seat memory, collective executive memory, and the formal evidence graph – with MemoBrain's native operations redefined in scientific semantics (Quarantine, Dialectical Fold, Perspective Recall).

We evaluate EpistemoBrain on **ForesightBlindspot**, a time-sliced evaluation protocol for blindspot discovery in computational social science, and report a controlled comparison of five baselines plus a six-variant ablation ladder on a scripted case with no model calls, no network, and no database, together with end-to-end verification on real research tasks. The comparison shows that blindspot coverage is driven by seat coverage – the single-agent baseline, after we corrected its seat-selection semantics (Section 4.3), finds one of seven gold blindspots while every seven-seat variant finds all seven – and the ablations show that removing the adversarial falsifier or the evidence auditor each costs a gold blindspot, while removing lineage inflates the system's own evidence-independence report. The comparisons also isolate precisely where the protocol's remaining contributions (memory, evidence gate) become observable under scripted gateways: on the blindspot metric they do not, which we report honestly as a property of the measurement, not a claim about the mechanisms.

Our contributions are threefold: (1) We introduce a *council protocol* for scientific controversy review in which seven specialized scientist agents interact through structured scientific actions (propose, support, challenge, qualify, fork, request, revise, dissent) under an explicit prohibition on majority voting, producing conditional consensus with preserved minority positions; (2) We design a *dual-graph evidence governance* architecture – process graph versus evidence graph, connected by an append-only, idempotent, replayable Scientific Event Ledger, a six-stage Evidence Gate, and a single Graph Projector whose write privileges are enforced at the database privilege level – together with a layered memory model that redefines executive memory operations in scientific semantics; (3) We provide **Poliscope**, an open-source implementation, and the **ForesightBlindspot** evaluation protocol with a five-baseline comparison and end-to-end verification, establishing a measurement basis for blindspot discovery in computational social science.

2 Related Work

2.1 Multi-Agent Debate and Collaboration

Multi-agent debate improves reasoning by having agents critique one another's answers (Du et al., 2023; Liang et al., 2023; Chen et al., 2023; Qian et al., 2024), and role-specialized panels have been proposed for domain reasoning. These systems typically end in a summarization step that aggregates positions; none addresses what this work considers the central failure mode: the summarizer can flatten a genuine dissent, and the agents can converge on a shared evidence error that looks like independent verification. EpistemoBrain differs on both axes: it forbids majority voting as a truth mechanism, and it subjects evidence – not arguments – to a six-stage gate before anything enters the formal knowledge graph.

2.2 Memory for Long-Horizon Agents

Long-horizon reasoning strains bounded contexts, motivating explicit memory architectures (Packer et al., 2023; Sumers et al., 2023). MemoBrain introduces executive memory for tool-augmented agents: a dependency-aware graph of reasoning steps with Flush/Fold/Recall operations that prune invalid steps, fold completed sub-trajectories, and preserve a compact, high-salience reasoning backbone under a fixed context budget (Qian et al., 2026). EpistemoBrain builds on MemoBrain as its executive-memory foundation but changes the unit of analysis: MemoBrain manages *one* agent's trajectory, while EpistemoBrain manages a *community*'s shared cognitive state. This forces three redefinitions – Flush becomes Quarantine (refuted claims change status, they do not disappear), Fold becomes Dialectical Fold (a dispute compresses only after preserving both strongest positions, hinge variables, and falsification conditions), and Recall becomes Perspective Recall (each seat recalls a role-specific projection of the shared evidence graph). Process memory and formal evidence are additionally separated into two graphs, because a scientist's momentary guess must never quietly upgrade into a formal conclusion.

2.3 Retrieval-Augmented Generation and Factual Verification

RAG grounds generation in retrieved documents (Lewis et al., 2020), and verification pipelines check claims against evidence (Thorne et al., 2018; Wadden et al., 2020; Min et al., 2023; Dhuliawala et al., 2023). These systems verify *textual entailment*; EpistemoBrain audits *scientific claims*, which requires several checks standard entailment systems do not perform: evidence levels (full text with exact quotes versus abstract-only metadata), citation entailment (an introduction hypothesis must not be reported as a tested result), causal upgrade policy (cross-sectional correlation must not become causal claims), evidence independence (papers sharing a dataset count as one independent evidence cluster), and source provenance (retraction status, preprint lineage). We reuse the production rules for these audits in our scoring functions rather than re-deriving parallel approximations.

2.4 LLM Agents for Science

LLM agents are increasingly applied to scientific tasks, from hypothesis generation to literature analysis. A recurring concern is that agentic systems can produce authoritative-sounding claims without grounding (Ioannidis, 2005; Open Science Collaboration, 2015); our design responds by making grounding a governance property rather than a prompt property. We are not aware of a system that combines a seven-scientist council protocol, dual-graph governance, an append-only event ledger with database-enforced write isolation, and a time-sliced blindspot evaluation benchmark in a single open-source implementation; Poliscope is, to our knowledge, the first.

3 Method

3.1 Overview

Figure 1 gives the architecture. A researcher submits a Research Contract (question, scope, budget, optional knowledge base and PDFs). The system proposes atomic claims; the researcher confirms which to investigate. A worker then runs the seven-seat council through eight phases, appending every formal change as an event to the Scientific Event Ledger. The Graph Projector – the ledger's single consumer and the evidence graph's single writer – applies the six-stage Evidence Gate to each candidate and materializes admitted nodes and edges. Gaps in the evidence graph generate Blindspot bounties that are scored, ranked, and broadcast back into the council (epistemic routing). At the end, the researcher receives a Research Brief, an interactive controversy map, an audit trail, and a synthesized final paper.

3.2 The Seven-Scientist Council

Each research task runs all seven seats. Table 1 lists the seats, the question each one owns, and its primary responsibilities. Each seat has an independent role specification, independent private state and private memory, and a different evidence-projection

Researcher → Council (7 seats × 8 phases) → Event Ledger → Evidence Gate → Graph Projector → Evidence Graph → Blindspot bounties → Council (loop)

Figure 1: EpistemoBrain architecture: council, ledger, gate, projector, and epistemic routing form a closed loop. Process memory (left, per-seat MemoBrain) and the formal evidence graph (right) are physically separated.

Seat	**Core responsibility**
Theory Builder	mechanisms and competing explanations
Causal Scientist	confounding, reverse causation, selection
Measurement Scientist	operationalization, self-report bias
Replication Scientist	power, sensitivity, replication history
Boundary Scientist	population, culture, platform limits
Adversarial Falsifier	strongest-attack counterexamples
Evidence Auditor	provenance, entailment, independence

Table 1: The seven scientist seats and their primary responsibilities.

weighting; the seats gather evidence without seeing one another, commit positions first, and only then exchange evidence, so cross-examination is genuinely cross-examination rather than restatement.

Scientists submit only structured actions: `PROPOSE`, `SUPPORT`, `CHALLENGE`, `QUALIFY`, `FORK`, `REQUEST`, `REVISE`, `DISSENT`; when challenged they may `DEFEND`, `REVISE`, `NARROW`, `WITHDRAW`, or `DISSENT`. Every action carries an actor, target, statement, evidence references, confidence, falsification condition, and novelty. Factual assertions without evidence, target-less comments, duplicated positions, and consent without new information are rejected by the ledger.

The protocol runs seven rounds: (1) *Independent Precommitment* – seats commit initial judgments and confidence levels without seeing one another (sealed first, read later); (2) *Professional Acquisition* – seats issue evidence requests from their own perspectives, merged by a query planner and executed against shared sources (OpenAlex, Crossref, Semantic Scholar, Unpaywall, and user-provided materials); (3) *Evidence Exchange* – structured evidence, not private reasoning histories, is exchanged; (4) *Cross-Examination* – targeted challenges on causality, measurement, statistics, boundaries, sources, and hidden assumptions; (5) *Blindspot Bounty* – gaps exposed by the evidence graph are scored on impact, uncertainty, investigability, novelty, and cost, ranked, and claimed; (6) *Joint Modeling* – a conditional consensus is drafted with shared understanding, strongest supporting and opposing positions, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions; (7) *Final Rejudgment* – seats judge independently again, recording belief updates and named dissents.

No majority voting decides scientific truth. The joint-modeling round refuses to produce a consensus when the strongest opposing position or a falsification condition is missing, and a seat that cannot accept the consensus receives a Dissent Certificate – a first-class node in the evidence graph – recording what they oppose, why, and what evidence would retract the dissent.

3.3 Layered Memory and Dialectical Operations

Memory is organized in three layers. *Private memory* holds each scientist's tool calls, failed routes, private hypotheses, and plans. *Collective executive memory* records structured cognitive events of the community: who proposed what, who attacked whom, which controversies remain open, which blindspots nobody has claimed. *The evidence graph* holds the audited formal knowledge: atomic claims, study findings, sources, constructs, contexts, blindspots, debate capsules, and dissent certificates.

MemoBrain's native operations are redefined for the scientific setting. *Quarantine* (from Flush): a refuted claim never disappears; it moves through `PROPOSED` → `SUPPORTED` → `CONTESTED` → `QUARANTINED` → `RESURRECTED/REJECTED`, and the record keeps who refuted it, what evidence is missing, and what conditions would resurrect it. *Dialectical Fold* (from Fold): a dispute folds into a Debate Capsule only after preserving shared understanding, the strongest supporting evidence, the strongest opposing evidence, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions – the original nodes remain and the capsule links back to them. *Perspective Recall* (from Recall): each seat's context is $\text{Context}_i = \text{ProcessRecall}_i + \text{EvidenceProjection}_i$, where the projection is a role-specific slice of the shared evidence graph, so the seven seats never read the same undifferentiated summary. Two further dynamics complete the set: *Fork* creates parallel research paths when evidence cannot distinguish competing explanations, *Resurrect* re-activates a quarantined hypothesis when new evidence meets its resurrection conditions, and *Merge* records conditional merge candidates for the researcher to adjudicate.

3.4 Dual-Graph Governance

Process and conclusion are stored apart. The *Process Graph* answers "how did the scientists work": tasks, tool calls, failed routes, challenges, and decisions, managed by MemoBrain and freely foldable. The *Evidence Graph* answers "what do we now know": ten node types (ResearchQuestion, Claim, Source, StudyFinding, Construct, Context, Blindspot, DebateCapsule, DiscriminatingStudy, DissentCertificate) and twelve edge types (SUPPORTS, REFUTES, QUALIFIES, CONTRADICTS, CONFOUNDS, MEDIATES, MODERATES, OPERATIONALIZES, DERIVED_FROM, APPLIES_IN, EXPOSES, TESTS).

The two graphs are coupled by the *Scientific Event Ledger*: every formal change is first appended as an idempotent, replayable, auditable event, and the *Graph Projector* – the ledger's only consumer and the evidence graph's only writer – applies the Evidence Gate and materializes admitted nodes and edges. The write isolation is enforced at the database privilege level, not by coding discipline: the application role holds no insert, update, or truncate privilege on the evidence-graph tables, and the projector role holds no privilege on the ledger, so a bug in either side cannot forge the other's data. Refuted, quarantined, and folded nodes are never physically deleted; they change status and remain traceable.

3.5 The Evidence Gate

Every candidate event passes six stages: *Schema validation*, *semantic deduplication*, *source verification*, *citation entailment*, *method quality*, and *graph consistency*. Evidence is graded A–D: Level A requires full text with exact quotes and precise location; Level B is abstract-plus-metadata; Level C is second-hand descriptions in reviews; Level D is web or news leads. Levels B/C/D can never alone support a high-confidence causal conclusion. Three audits are applied to formal findings: source authenticity (DOI, title, authors, version, retraction, PDF identity), citation entailment (does the quote support the claim; is correlation reported as causation; are qualifiers dropped), and method quality (directness, design strength, measurement validity, statistical precision, replicability, external validity). A *CausalUpgradePolicy* intercepts the combination of cross-sectional design plus causal claim at stage six and quarantines the event with an audit record. The pipeline also tracks evidence independence: lineage detection identifies same-dataset, overlapping-sample, preprint, extension, reanalysis, and meta-analysis-inclusion dependencies, and the interface reports both the paper count and the independent-evidence-cluster count – six papers sharing one dataset count as one independent evidence cluster.

Four mechanisms defend against the subtler failure of *shared* evidence errors: blind evidence review (prestige signals such as author, journal, and citation counts are stripped from first substantive judgments), independent dual extraction (high-impact findings are extracted twice through separate paths and discrepancies are escalated), source diversity constraints (when all supporting evidence for a claim shares a dataset, team, country, or design, a Blindspot node is automatically generated), and adversarial retrieval (six classes of reverse queries – refutation, null results, alternative theories, measurement criticism, replication failure, boundary inversion – are executed against the evidence sources for each confirmed claim).

3.6 Epistemic Routing and Blindspot Bounty

The council is not a scripted run of seven rounds. Gaps exposed by the evidence graph generate Blindspot bounties, scored on impact, uncertainty, investigability, novelty, and normalized cost, ranked, and broadcast for the seats to claim – the evidence state drives the next investigation direction rather than a moderator reading from a script. Blindspot is therefore a first-class object in the product (a radar view ranked by impact and investigability), not a limitations paragraph. When the bounty round closes, the system may also propose a *DiscriminatingStudy*: a study design with competing predictions that would resolve an open conflict and the expected information gain.

3.7 Human Directional Checkpoint

After the blindspot bounty and before joint modeling, the task pauses at a single, fixed checkpoint (`AWAITING_COUNCIL_INPUT`). The researcher may review the seven seats' precommitment positions and submit a short directional note (e.g., "prioritize cross-cultural boundary conditions") or explicitly leave it blank and continue; both are equally valid. The note is injected into joint modeling only as an explicitly labeled "researcher directional note, not a scientific judgment"; it never changes Evidence Gate decisions, never serves as evidence for or against any claim, and never enters Debate Capsules or Dissent Certificates. The same evidence input must produce identical claim admission, dissents, and gate decisions whether or not a note was provided – a property asserted by tests. The checkpoint is directional control, not an eighth vote.

3.8 Engineering Realization

Poliscope is a modular monolith with an asynchronous worker, PostgreSQL as the source of truth, and no workflow state living only in memory. The worker claims tasks with `SELECT ... FOR UPDATE SKIP LOCKED`; a council run executes each phase in a single transaction as the application role, then commits, and the projector runs in a second transaction as the projector role – two identities, two transactions, one ledger. All model calls go through a Model Gateway (OpenAI-compatible endpoints, streaming with non-streaming fallback, per-task model configuration, cost and latency accounting); all external tools go through a Tool Gateway (OpenAlex, Crossref, Semantic Scholar, Unpaywall, user knowledge bases). Single-seat failure degrades the round (the seat is marked absent with a reason on the ledger) instead of aborting the task; structured-output repair failures are quarantined and never write the formal graph. Live views are served as two SSE streams – the formal ledger stream and a best-effort process stream for token-level progress – so the audit trail is never polluted by reasoning noise. The implementation is open-source (MIT) at https://github.com/Fishman-free/poliscope.

4 Experiments

4.1 The ForesightBlindspot Benchmark

We design **ForesightBlindspot**, a time-sliced evaluation protocol for blindspot discovery in computational social science (domain: digital behavior, social media, and mental health). Each case is a contested question with a cutoff date: the system may read only the pre-cutoff corpus, and gold blindspots are gaps that later high-quality research confirmed, had reasonable premonitory evidence before the cutoff, and do not depend on entirely unforeseeable platforms, events, or methods. Leakage controls are built into the protocol: closed pre-cutoff corpora, no future citations, anonymizable variables, and an explicit disclosure that base-model parameters may contain future knowledge – a risk that cannot be fully eliminated. The metric family covers blindspots (recall, precision, high-impact recall, specificity, actionability, foresight distance), evidence (citation existence, citation entailment, evidence independence, causal overclaim rate, full-text coverage), memory (compression ratio, scientific backbone retention, contradiction retention, long-horizon drift, recovery quality), collaboration (role diversity, debate utility, belief-update quality, dissent preservation, false consensus rate), and cost (cost per valid blindspot, tokens per admitted finding).

4.2 Baselines

Five baselines are configurations of the real council rather than separate implementations, so each adds exactly one capability over the previous: (1) *Single-Agent Deep Research* – one researcher (the theory builder) with no protocol, no memory, and no gate; (2) *Fixed Multi-Agent Debate* – seven seats with an undifferentiated deliberator, no per-seat memory, no gate; (3) *Council + Linear Context* – seven seats sharing one undifferentiated transcript; (4) *Council + MemoBrain without Evidence Engine* – seven seats with per-seat memory but the gate removed; (5) *Full Poliscope* – the complete system.

4.3 Baseline Construction Correctness

Before reporting numbers, we validated that the baselines are what they claim to be – and found two construction defects that the first evaluation run exposed. First, the single-agent baseline selected its seat by the alphabetically first seat of an ordered enumeration, which happened to be the adversarial falsifier: the "lone researcher" was actually the falsifier, and in our scripted material the falsifier is the seat with the strongest blindspot answers, so the single agent scored as high as the full council (F1 0.8 on all five variants). Second, the final-rejudgment handler iterated all seven seats unconditionally, minting placeholder judgments for seats that never participated. We corrected both: the single-agent baseline explicitly uses the theory builder, and the rejudgment handler respects the participating-seat set with a compatible fallback so the seven-seat production path is byte-for-

Baseline	Rec.	Prec.	F1
Single-Agent Deep Research	0.143	0.500	0.222
Fixed Multi-Agent Debate	1.000	0.778	0.875
Council + Linear Context	1.000	0.778	0.875
Council + MemoBrain, no gate	1.000	0.778	0.875
Full Poliscope	1.000	0.778	0.875

Table 2: Blindspot recall, precision, and F1 on the scripted demo case (B_dev). All four seven-seat variants find all seven gold blindspots; the single agent, restricted to the theory builder, finds one.

Variant	Rec.	Prec.	F1	Indep.	Unfilled
Full Poliscope	1.000	0.778	0.875	0.667	1
− Precommitment	1.000	0.778	0.875	0.667	1
− Adversarial falsifier	0.857	0.857	0.857	0.667	1
− Evidence auditor	0.857	0.750	0.800	0.667	1
− Dialectical fold	1.000	0.778	0.875	0.667	2
− Lineage	1.000	0.778	0.875	1.000	1
− MemoBrain	1.000	0.778	0.875	0.667	1

Table 3: Ablations of the full system on the same scripted case (B_dev). "Indep." is evidence independence; "Unfilled" counts slots the run records as unfinished rather than fabricating. Each ablation removes exactly one capability; the adversarial falsifier and the evidence auditor each carry a gold blindspot of their own, which is why removing them costs recall.

byte unchanged. A regression guard now asserts that single-seat blindspot coverage is strictly below multi-seat coverage, so a future regression that collapses the baseline ladder fails immediately. We report this correction process because the measurement instrument's construction is part of the measurement.

4.4 Results on the Scripted Case

Table 2 reports the five baselines on a scripted demo case (screen time and adolescent depressive symptoms) in which every gateway is deterministic – no model calls, no network, no database – and each seat nominates one specialist blindspot in the bounty round (the falsifier nominates two, one deliberately duplicating the boundary scientist's nomination to model overlapping multi-seat nominations). Gold blindspots are one keyword per specialist seat.

Three findings stand out. First, blindspot coverage is driven by *seat coverage*: the lone researcher finds one of seven specialist blindspots (F1 0.222) while every seven-seat variant finds all seven (F1 0.875) – the council's contribution is real and large, and it is attributable to the seats rather than to any single seat's presence. Second, the distinguishing factor is the *set of participating seats*, not the presence of any particular seat: after the construction fix, the single agent's score is determined by what its own seat contributes, not by whether the falsifier happens to be present. Third – reported honestly as a property of the measurement – the four seven-seat variants score identically on the blindspot metric under scripted gateways. This is a structural consequence of the harness: every variant collects all seven seats' outputs, the scripted gateway ignores prompts and memory, and the bounty events are admitted unconditionally by the gate. The contributions of the protocol, memory, and evidence gate to blindspot *integration* are not observable through this metric on scripted material; they surface in the auxiliary scores (dissent preservation, citation entailment, evidence independence) and require real-model runs to observe on blindspots. We treat this as an identified boundary of the measurement, not a claim about the mechanisms.

4.5 Ablation Analysis

Beyond the five-baseline ladder, we ablate the full system one capability at a time (design spec 11.4): six *full minus X* variants, each removing exactly one mechanism while keeping everything else – including the evidence gate – intact. The removals are (1) independent precommitment (the phase does not run); (2) the adversarial falsifier seat; (3) the evidence auditor seat; (4) the dialectical fold (JOINT MODELING skips the DebateCapsule and records an unfilled slot instead of silently dropping the opposition); (5) evidence lineage (acquired sources arrive without dataset metadata, so evidence independence cannot see that two papers share one dataset); (6) MemoBrain (per-seat private memory replaced by one undifferentiated transcript). Each ablation is a configuration of the same orchestrator the full system runs – the switch is one flag or one seat subset, not a reimplementation – so the contrast with Full Poliscope isolates that capability's contribution.

Table 3 shows three mechanism-level effects. First, removing the adversarial falsifier drops recall from 1.000 to 0.857: the publication-bias blindspot was *its* nomination and no other seat covers it, so a council without a dedicated falsifier misses exactly the gap it exists to find. Precision simultaneously rises (0.778 to 0.857) because the falsifier's second nomination deliberately duplicates the boundary scientist's – without the falsifier, no statement is double-paid. Second, removing the evidence auditor costs recall as well (1.000 to 0.857, F1 0.800): the provenance blindspot belonged to that seat. Third, removing lineage raises evidence independence from 0.667 to 1.000 – without dataset metadata, two papers that share a cohort are counted as independent evidence (CLAUDE.md 7.4's paper-count-vs- independent-cluster distinction), so the number the system reports about its own evidence quality is *inflated by the absence of the mechanism that would correct it*. The dialectical-fold ablation cannot affect the blindspot metric but is visible in the honesty signal: the DebateCapsule disappears and the run records one additional unfilled slot rather than pretending the debate had no opposition – dissent preservation is a property of the protocol, and its absence is reported, not hidden.

Two ablations show no effect on this scripted material, and we report that plainly rather than manufacturing a difference: removing precommitment removes the seven PRECOMMITMENT_SEALED events but leaves every score unchanged, and removing MemoBrain changes nothing either. This is the same structural boundary identified for the baseline ladder: the scripted gateway answers identically regardless of prompt, memory, or phase history, so protocol-level integration effects – what precommitment changes about later reasoning, what per-seat memory changes about what a seat recalls – are not observable without real models. The two ablations that do show effects (*falsifier*, *auditor*) do so because they remove a *participant* with scripted content, not because the mechanism's behavioral consequences are being measured. The *lineage* ablation shows an effect for the same reason: the strip changes the data the scoring function sees. We therefore treat the three effects as validated mechanism checks and the two null results as an identified measurement boundary, not as evidence that the two capabilities contribute nothing.

4.6 Results with a Real Model and a Semantic Judge

The scripted case is deterministic by construction, and its blindspot metric rests on a *keyword* proxy: "reverse_causation" matches only statements that literally contain those words (underscore-separated and case-folded). The scripted gateway writes its statements to contain the keyword, so the proxy works there. It does not survive contact with a real model: free-form output such as "the direction of effect could run the other way" names the reverse-causation gap without containing the token, so the keyword score reports recall 0.0 even when the gap was found. To measure blindspot quality under a real model we added a *semantic judge*: for each gold concept, the same model gateway (no second vendor SDK) is asked which numbered candidate statements semantically address that concept, and recall/precision are derived with the identical definitions the keyword score uses. A judge that is unreachable or whose structured answer is quarantined returns *unmeasured* rather than silently 0.0.

We then ran five key variants (the full system, the single-agent baseline, and the three mechanism ablations whose scripted effects were largest) through the real gateway (DeepSeek-V4-Flash on the official DeepSeek API), three runs each, and scored blindspots with the semantic judge. The initial eleven-variant single-run sweep was abandoned because the first endpoint rate-limited for two days, polluting twelve runs (unfilled slots jumped to 46 and the judge was unreachable for every semantic call); those runs were discarded rather than reported. Table 4 reports the three-run means (F1 recomputed from the mean recall and precision, not the mean of per-run F1s).

Three findings follow. *First*, the semantic judge repairs the measurement: the keyword proxy reports recall 0.0 for the full system on free-form text, while the semantic judge produces a readable, non-degenerate distribution. *Second*, the repeated runs turn the earlier single-run reading – "sampling variance dominates" – into a stable, reproducible *precision–recall trade-off*: full Poliscope achieves perfect precision (1.000) at the cost of the lowest recall (0.286, exactly two of seven gold concepts in all three runs), while every simplified variant – including the lone researcher – finds nearly all gold concepts (recall 0.809–1.000) with markedly lower precision (0.406–0.586). The precommitment removal reproduces recall 1.000 in every run, so the earlier single-run observation was not a sampling artifact but a real, repeated effect. Unfilled slots explain the gap: the full system leaves 12.3 slots unfilled on average (the model produced no structured nomination), versus 1.0–1.7 for the simplified variants. Protocol burden on a flash-tier model is the binding constraint: the council's governance mechanisms buy precision at the cost of recall, and under a weak model the recall cost dominates F1. *Third*, the deterministic data-level scores (citation entailment 0.5, evidence independence 0.667, dis-

Variant	Sem-R	Sem-P	Sem-F1	Unf.	n
Single-Agent Deep Research	0.809	0.544	0.651	1.0	3
Full Poliscope	0.286	1.000	0.444	12.3	3
− Precommitment	1.000	0.490	0.658	1.0	3
− Adversarial falsifier	1.000	0.406	0.577	1.7	3
− Evidence auditor	0.952	0.586	0.726	1.3	3

Table 4: Blindspot scoring on repeated real-model runs, means over $n = 3$. "Sem" is the semantic judge; R/P/F1 are recall, precision, and F1; "Unf." is the mean number of unfinished evidence slots. The keyword proxy is omitted: on this endpoint it again collapses to 0 for the full system, and the semantic judge is the controlled measurement.

sent preservation 1.0) are identical across all fifteen runs, as expected for scoring that does not depend on model output.

We report this honestly rather than as a positive result. The semantic judge is a usable measurement instrument for blindspot quality – it replaces a keyword proxy that demonstrably fails on free-form text – but the mechanisms EpistemoBrain adds are *integration-level*: they shape what a seat reasons and recalls across phases, and observing their benefit requires a model strong enough to sustain the full protocol without the seat absence seen here (or a lighter protocol). That is an engineering follow-on, not a silent assumption; the scripted case already isolates each mechanism's structural contribution, and this subsection records the real-model boundary at which that isolation stops.

4.7 System Verification

Beyond the controlled comparison, Poliscope is verified end to end. The test suite comprises 752 unit tests and 364 integration tests over a real PostgreSQL instance with three database identities, including privilege tests that assert the application role cannot write the evidence graph and the projector role cannot rewrite the ledger – the dual-graph isolation is a database fact, not a code convention. Real research tasks run through the full seven-seat council on real OpenAI- compatible endpoints, producing conditional consensus, Debate Capsules, Dissent Certificates, and a synthesized final paper with references bound to DOIs; tasks that exhaust budget or lose seats finish as "completed with gaps" with every absent seat and unfilled slot named in the report, never as a fabricated success. The event ledger is idempotent and replay-safe (re-running a task is a no-op), and the streaming live view survives reconnection via Last-Event-ID.

5 Conclusion

We presented EpistemoBrain, the organizing brain of the open-source Poliscope system: a seven-scientist council with a formal protocol, layered memory, dual-graph evidence governance, an append-only event ledger with database-enforced write isolation, an evidence gate with three-layer audit, and epistemic routing through blindspot bounties. The design's thesis is that reliable blindspot discovery is an organizational and governance problem, not a prompt problem: dissent must survive by construction, evidence must be audited by construction, and process must never become evidence by construction. The ForesightBlindspot protocol, the five-baseline comparison, and the six-variant ablation ladder establish a measurement basis – with the caveat, reported openly, that scripted gateways make the protocol's integration-level contributions unobservable on the blindspot metric (two of the six ablations are null on this material for exactly that reason). A real-model run further shows that a keyword proxy for blindspot recall fails on free-form text and that a semantic judge repairs the measurement. Repeated runs then expose a stable boundary rather than noise: under a flash-tier model the full protocol's governance buys perfect blindspot precision at the cost of recall – every simplified variant, including a lone researcher, finds more gold concepts – with seat absence (12.3 unfilled slots versus 1.0–1.7) as the binding constraint. Observing integration-level benefits thus requires either a stronger model or a lighter protocol; both are honest engineering follow-ons. A time-sliced corpus with a held-out split and human annotation remains the next step of this research program.

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A Appendix: Implementation Notes

Deterministic evaluation harness. The Foresight-Blindspot harness (`packages/evaluation/`) runs the five baselines and the six ablation variants as configurations of the real orchestrator against a database-less event sink, so a run is deterministic and fast enough for unit tests; scoring functions reuse the production audit rules (causal upgrade policy, citation entailment verifier, lineage detection and evidence clustering) instead of re-deriving them, so a variant’s score reflects the same standard the gated system is held to. Each ablation is one protocol switch on that same orchestrator – a phase subset, a seat subset, a fold flag, or a lineage-free acquirer – never a reimplementation (Section 4.5). Gold blindspot matching is currently a keyword proxy for human annotation, which does not yet exist; the agreement skeleton (Kappa/Alpha statistics, double annotation, adjudication) is implemented and awaiting annotation data.

Permissions as design. PostgreSQL grants implement the write isolation: the application role can append to the ledger and read the evidence graph but cannot insert, update, or truncate graph tables; the projector role can write the graph and advance event status but cannot rewrite event payloads; neither role can truncate anything. A documented exception grants the application role DELETE for the session-lifecycle operation of destroying a whole task at the researcher’s explicit request – the one path by which evidence rows may be physically removed.

Reproducibility. The repository pins migrations (Alembic), seeds, and Docker Compose services (postgres, migrate, api, worker, web, caddy); the demonstration task seed runs the real orchestrator with scripted model and tool stand-ins, and the evaluation harness runs with scripted gateways, so all numbers in this paper are reproducible without external credentials.